

# The DobbertinLego library

Two bricks **F** **L**, one connective **T**  $\rightarrow$  Dobbertin's Theorem 1, step (1)  $\Rightarrow$  (2)

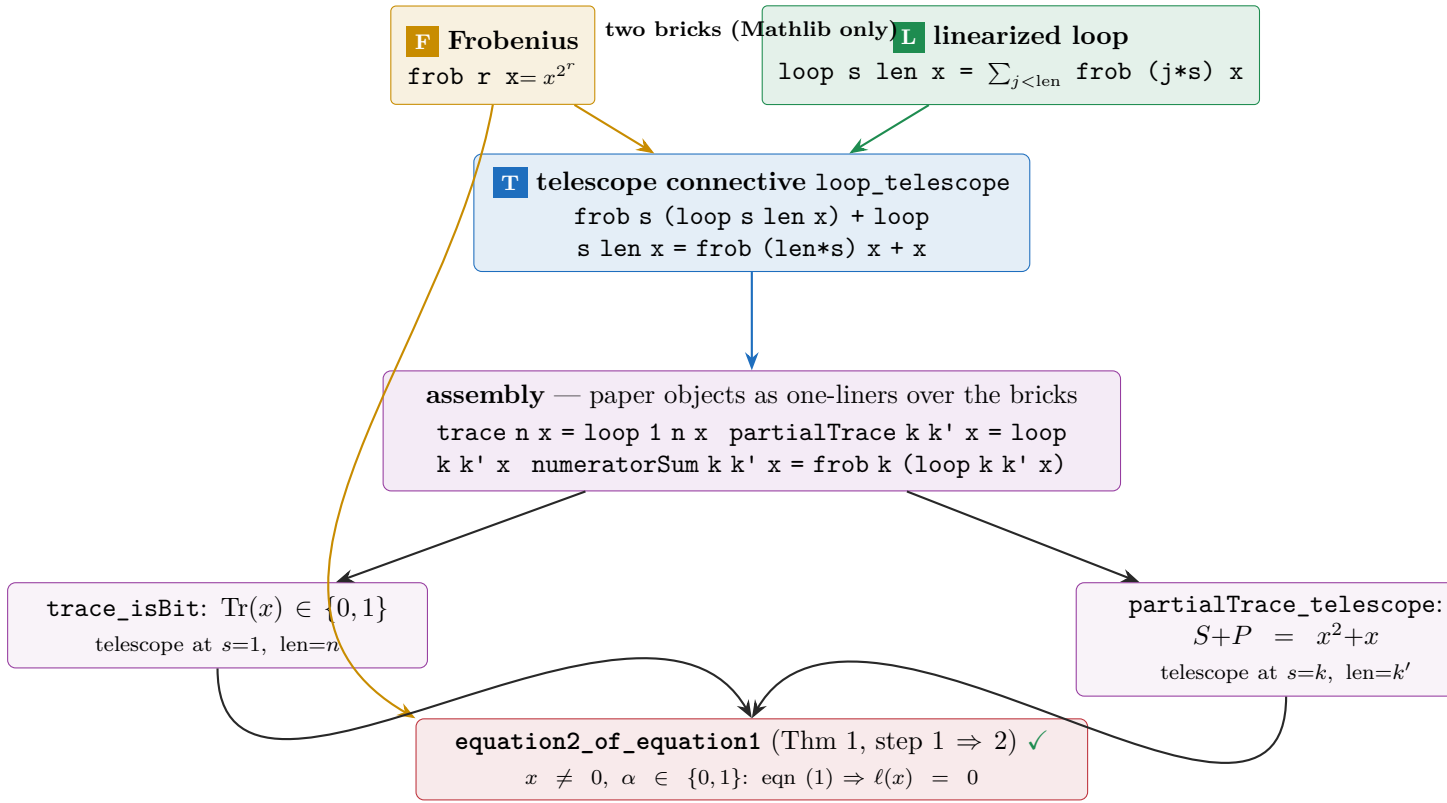
A picture book of the minimal DobbertinLego Lean library — `import DobbertinLego`

**What this is.** A deliberately minimal, end-to-end rebuild of the first step of the proof of Theorem 1 in Dobbertin's 1999 paper *Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets* — the implication equation (1)  $\Rightarrow$  equation (2). The point of this version is that the *entire* proof snaps together from just **two primitive bricks** and **one connective**:

**F** Frobenius `frob` **L** linearized loop `loop` **T** telescope `loop_telescope`.

Every box drawn below is a  $\checkmark$  sorry-free Lean declaration; the headline depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`. The four modules are `DobbertinLego/Frobenius`, `DobbertinLego/Loop`, `DobbertinLego/Assembly` and the root `DobbertinLego`.

## The whole library on one page



The dependency edges are literal Lean imports. The three objects `trace`, `partialTrace`, `numeratorSum` — three separate definitions in the earlier library — are now a *single* gadget **L** at different settings, and the two telescoping proofs collapse to *one* connective **T**.

## 1 Brick **F** — Frobenius (`DobbertinLego/Frobenius.lean`)

The doubling map on exponents,  $x \mapsto x^{2^r}$ . Its four laws are the entire finite-field vocabulary the proof needs.

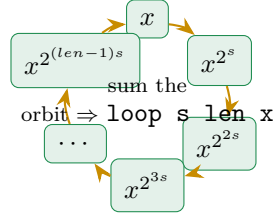
```

def frob (r : ℕ) (x : F) : F := x ^ (2 ^ r)
frob_add      : frob r (x + y) = frob r x + frob r y      ✓ -- additive
frob_comp     : frob a (frob b x) = frob (a + b) x        ✓ -- iterates compose
frob_periodic : |F|=2^n ⇒ frob r x = frob (r % n) x        ✓ -- period n
frob_card     : |F|=2^n ⇒ frob n x = x                    ✓ -- Fermat

```

## 2 Brick **L** + connective **T** (DobbertinLego/Loop.lean)

Gadget **F** closed into a loop of length  $\text{len}$  with step  $s$ : the running sum of the doubling map along an arithmetic progression of exponents.



```

def loop (s len : ℕ) (x : F) : F := ∑_j<len frob (j*s) x
loop_succ : loop s (len+1) x = loop s len x + frob (len*s) x      ✓
loop_telescope : frob s (loop s len x) + loop s len x              ✓ T
                = frob (len*s) x + x

```

The connective **T**. One induction on  $\text{len}$ : applying one more Frobenius step (brick **F**, additive) to the loop and adding the loop back makes every interior power appear twice, so it cancels in characteristic 2, leaving only the two endpoints. This single lemma is the paper’s “add the  $2^k$ -th power of this equation to itself”.

## 3 Assembly (DobbertinLego/Assembly.lean)

The paper’s objects are now one-liners; the two auxiliary facts are two uses of **T**.

| paper object                                      | LEGO expression     | Lean                                                    |
|---------------------------------------------------|---------------------|---------------------------------------------------------|
| trace $\text{Tr}(x) = \sum_{i<n} x^{2^i}$         | <b>L</b> at $s=1$   | <code>trace n x = loop 1 n x</code>                     |
| partial trace $P(x) = \sum_{j<k'} x^{2^{jk}}$     | <b>L</b> at $s=k$   | <code>partialTrace k k' x = loop k k' x</code>          |
| numerator sum $S(x) = \sum_{i=1}^{k'} x^{2^{ik}}$ | <b>F</b> ◦ <b>L</b> | <code>numeratorSum k k' x = frob k (loop k k' x)</code> |

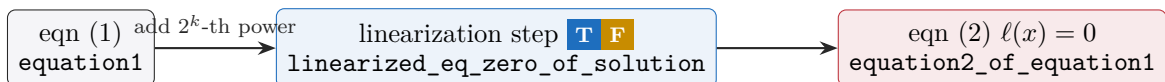
```

numeratorSum_eq_frob_partialTrace : numeratorSum k k' x = partialTrace k k' x ^ (2^k)  ✓ -- rfl
trace_isBit : |F|=2^n ⇒ trace n x = 0 ∨ trace n x = 1      ✓ -- T at s=1
partialTrace_telescope : |F|=2^n → k*k'%n=1 ⇒
    numeratorSum k k' x + partialTrace k k' x = x^2 + x    ✓ -- T at s=k

```

Two uses of one connective. `trace_isBit` is **T** at  $s=1$ ,  $\text{len}=n$ :  $\text{Tr}^2 + \text{Tr} = \text{frob } n x + x = 0$  (by `frob_card`), so  $\text{Tr}$  is idempotent, hence a bit. `partialTrace_telescope` is **T** at  $s=k$ ,  $\text{len}=k'$  with the endpoint  $\text{frob}(k'k)x$  collapsed to  $x^2$  by `frob_periodic` and  $k \cdot k' \equiv 1 \pmod{n}$ . And  $S = P^{2^k}$  holds *by definition* (`rfl`).

## 4 The headline (DobbertinLego.lean)



```

def equation1 (n k k' α : ℕ) (c x : F) : Prop :=
  c * x^(2^k+1) = numeratorSum k k' x + (α:F) * trace n x
def linearized (k : ℕ) (c x : F) : F :=
  c^(2^k) * x^(2*(2^k)) + x^(2^k) + c*x + 1

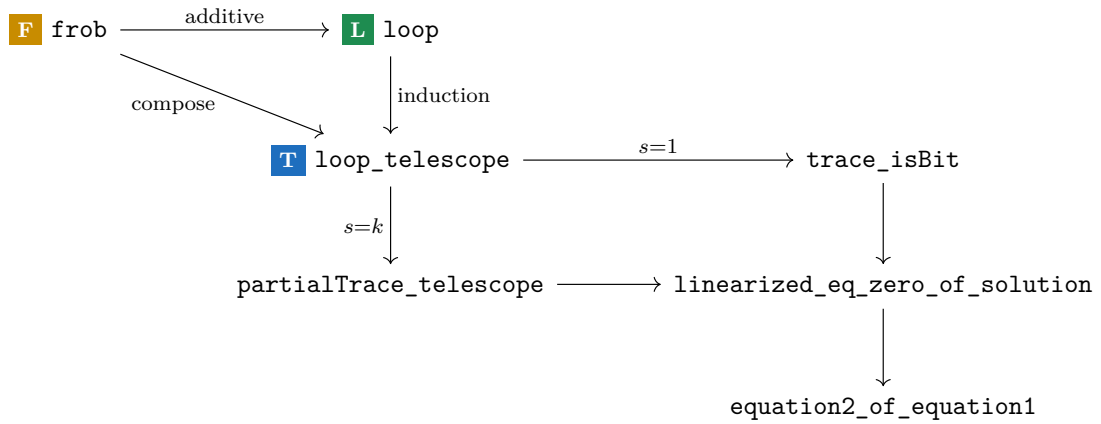
theorem equation2_of_equation1 (hn : |F| = 2^n) (hkk' : k*k' % n = 1)
  (hα : α=0 ∨ α=1) (hx : x ≠ 0) (h : equation1 n k k' α c x) :
  linearized k c x = 0

```

✓

The proof: collapse  $\alpha \cdot \text{Tr}(x)$  to a bit  $\varepsilon$  (via `trace_isBit`); solve (1) for  $P$  using  $S = P^{2^k}$  and the telescope  $S + P = x^2 + x$ ; raise to the  $2^k$  power (brick **F** additive,  $\varepsilon^{2^k} = \varepsilon$ ); cancel  $\varepsilon$  and divide by  $x^{2^k} \neq 0$  to read off  $\ell(x) = 0$ . *Why*  $x \neq 0$ . At  $x = 0$  the cleared (1) holds vacuously ( $0 = 0$ ) while  $\ell(0) = 1 \neq 0$ ; the hypothesis is genuinely required.

## Compositional build diagram



**One sentence.** The step  $(1) \Rightarrow (2)$  of Dobbertin’s Theorem 1 is *one* gadget — the linearized Frobenius loop **L** built on the doubling brick **F** — together with its *one* telescoping connective **T**; the paper’s trace, partial trace and numerator sum are that gadget at three settings, its “add the  $2^k$ -th power” move is that connective, and the headline is a short assembly on top. *Scope*: this covers the first step only; the two root-counting cases, Theorem 1 proper and the difference-set half are not part of this minimal library.